INCORRECT

$$\int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - 2\cos(\theta) + 2\cos^2(\theta)\right) d\theta$$

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$$\int_0^{\frac{\pi}{3}} (1 - 4\cos(\theta) + 4\cos^2(\theta))\, d\theta$$

CORRECT (SELECTED)

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INCORRECT

$$\int_0^{\frac{\pi}{2}} (1 - 4\cos(\theta) + 4\cos^2(\theta))\, d\theta$$

1 / 4

This is the formula for the area enclosed by a polar curve $r(\theta)$ between $\theta = \alpha$ and $\theta = \beta$:

$$\int_\alpha^\beta \frac{1}{2}(r(\theta))^2 d\theta$$

We know $r(\theta)$ but we still need to figure out α and β.

2 / 4

Notice that $r(0) = -1$, and indeed the curve starts at $(-1, 0)$. So $\alpha = 0$.

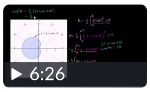
β is the θ-value in the range $0 \le \theta \le \pi$ for which $r(\beta) = 0$. So $\beta = \frac{\pi}{3}$.

Show me how you found β.

+ Get another hint (2/4)

Related content

Area bounded by polar curves



Worked example: Area enclosed by cardioid

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